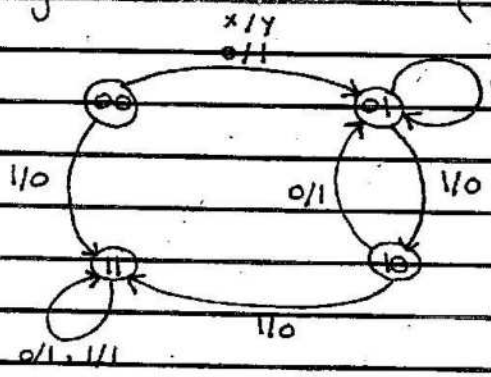
Logic Design / final
sequential circuits أمثلة
م. رغيده محمد
٢٠١٥/٥/١٤

أمثلة تصميم منطقي رقمي
Final حالة

⊗ أمثلة sequential circuits

① Trace the following state diagram, and draw the circuit using DFF (مثال المحاضرة مع الحل تفاعلي)



X: input
Y: output

⇒ we need 2FF.

الجدول ★

Present state A	Present state B	input X	Next state A	Next state B	out Y	FF inputs D_A	FF inputs D_B
0	0	0	0	1	1	0	1
0	0	1	1	1	0	1	1
0	1	0	0	1	0	0	1
0	1	1	1	0	0	1	0
1	0	0	0	1	1	0	1
1	0	1	1	1	0	1	1
1	1	0	1	1	1	1	1
1	1	1	1	1	1	1	1

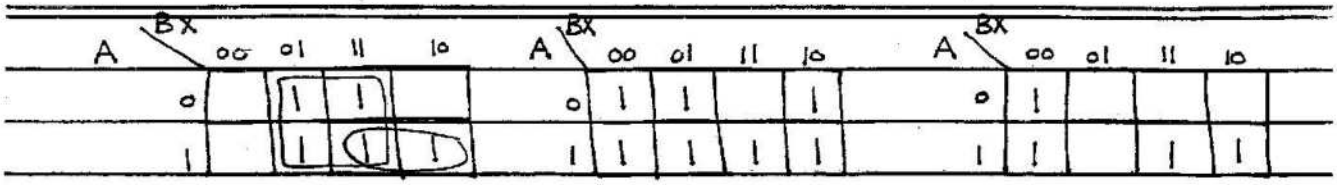
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نشتق معادلات Y ، D_A ، D_B ونختصرها بالـ K-map

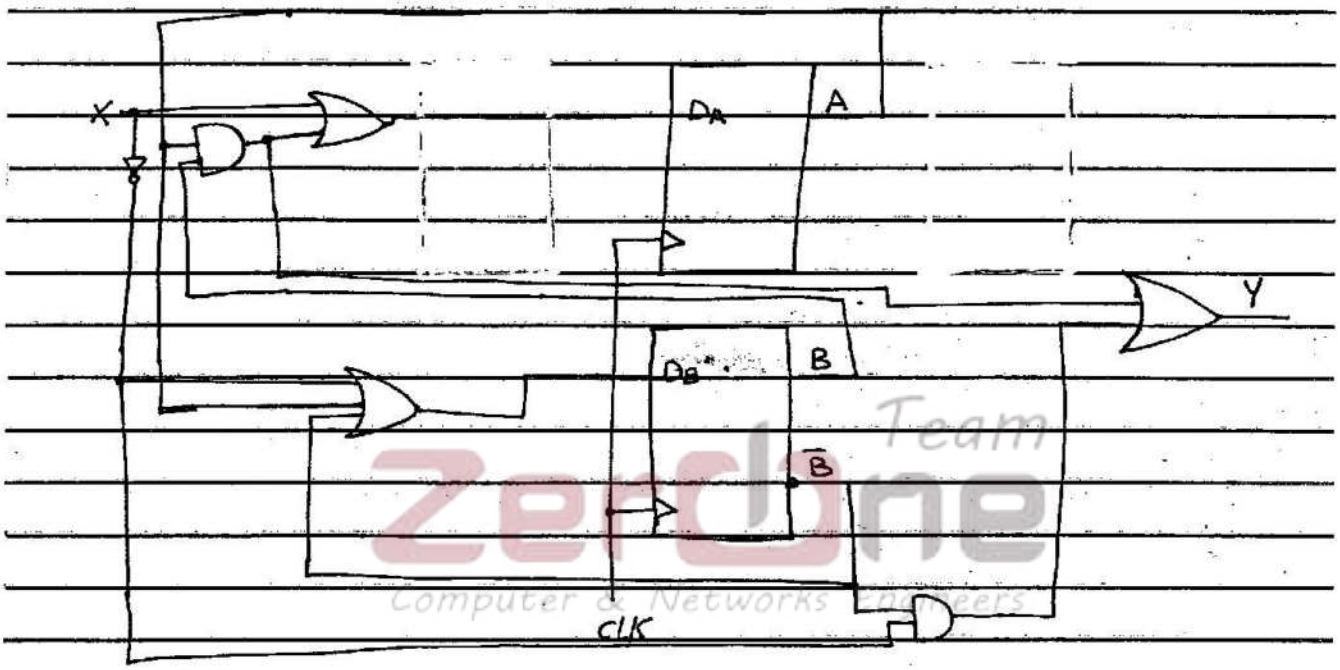
لأن الدائرة D ولأنه $Q(t+1) = D$

للمواجع D فإنه $A(t+1) = D_A$

$B(t+1) = D_B$



$D_A = X + AB$ $\quad D_B = \bar{B} + \bar{X} + A$ $\quad Y = \bar{B}\bar{X} + AB$



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$A(t+1) = D_A = X + AB$

$B(t+1) = D_B = \bar{B} + \bar{X} + A$

$Y(t) = \bar{B}\bar{X} + AB$

② the same as ① but with JK FF.

Present state A	B	Input X	Next state A	B	out Y	FF inputs J_A	K_A	J_B	K_B
0	0	0	0	1	1	0	X	1	X
0	0	1	1	1	0	1	X	1	X
0	1	0	0	1	0	0	X	X	0
0	1	1	1	0	0	1	X	X	1
1	0	0	0	1	1	X	1	1	X
1	0	1	1	1	0	X	0	1	X
1	1	0	1	1	1	X	0	X	0
1	1	1	1	1	1	X	0	X	0

A \ BX	00	01	11	10
0		1	1	
1	X	X	X	X

A \ BX	00	01	11	10
0	X	X	X	X
1	1	1		

$J_A = X$ $\quad K_A = \bar{B}\bar{X}$

A \ BX	00	01	11	10
0	1	1	X	X
1	1	1	X	X

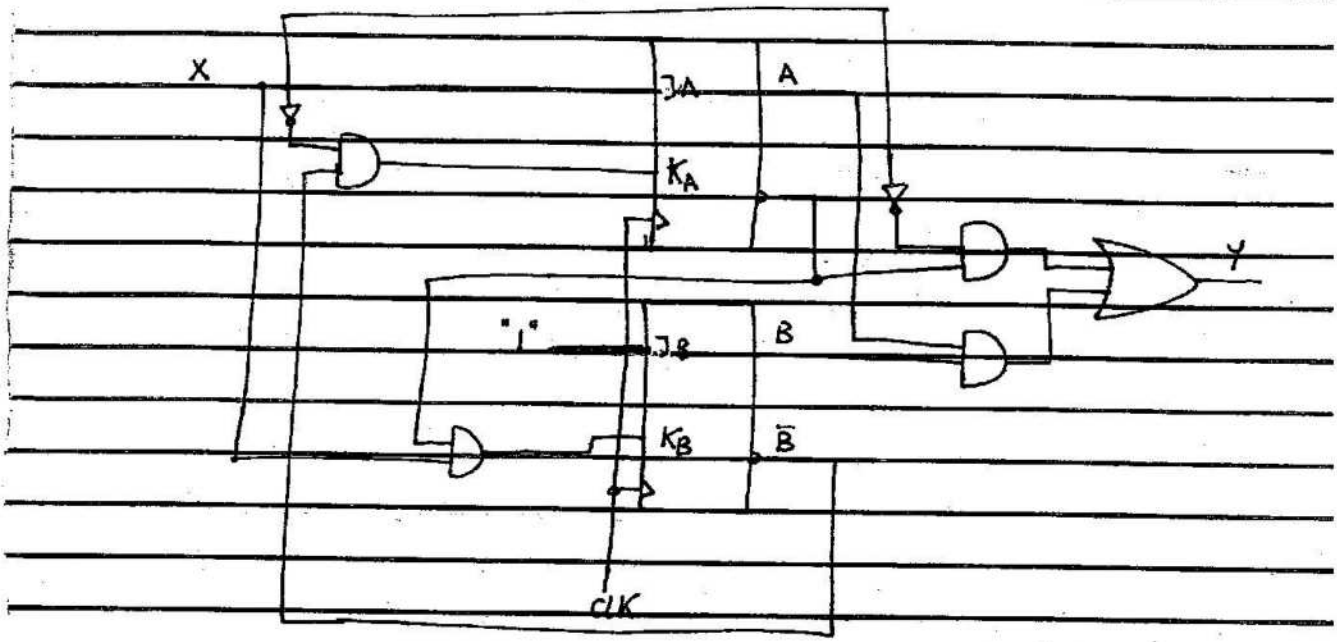
A \ BX	00	01	11	10
0	X	X	1	
1	X	X		

$J_B = 1$ $\quad K_B = \bar{A}X$

A \ BX	00	01	11	10
0	1			
1	1		1	1

$Y = \bar{B}\bar{X} + AB$

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* Char. equat.

$$Q(t+1) = J\bar{Q}(t) + \bar{K}Q(t)$$

$$A(t+1) = J_A \cdot \bar{A}(t) + \bar{K}_A \cdot A(t)$$
$$= X \cdot \bar{A}(t) + \overline{B\bar{X}} \cdot A$$
$$= X\bar{A} + A(B + X)$$

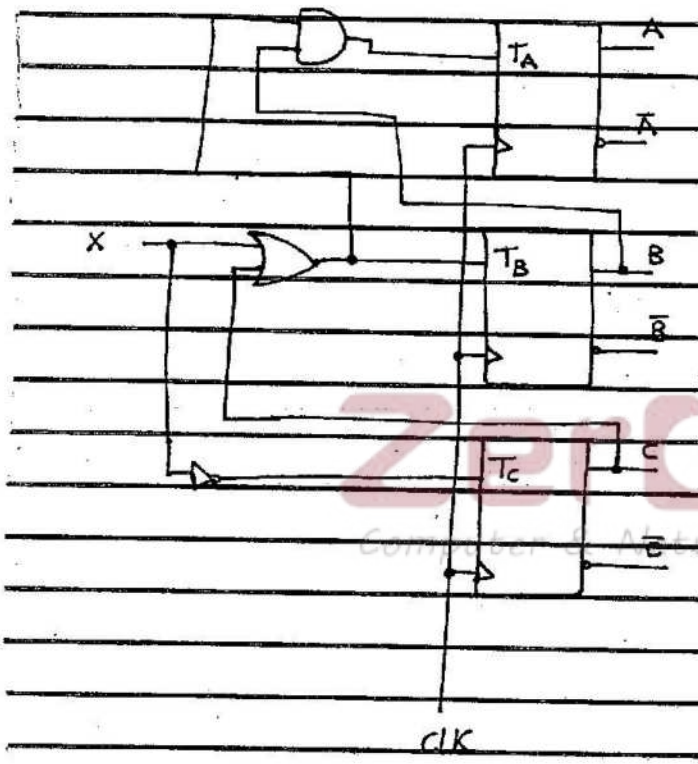
$$B(t+1) = 1 \cdot \bar{B}(t) + \overline{\bar{A}X} \cdot B(t)$$
$$= \bar{B} + B(A + \bar{X})$$

$$Y(t) = \bar{B}\bar{X} + AB$$

⊛ Try to solve the same example using:
1) D T FF
2) SR FF

③ for the sequential circuit shown below,

1- Derive the state table
2- Derive the state equations → analyze
3- Draw the state diagram.



$T_A = (X+C).B$

$T_B = X+C$

$T_C = \bar{X}$

$A(t+1) = T_A \oplus A$

$= (XB + BC) \oplus A$

$B(t+1) = T_B \oplus B$

$= (X+C) \oplus B$

$C(t+1) = T_C \oplus C$

$= \bar{X} \oplus C$

سوف لدينا 3 مراجع ومدخل واحد X ← بذا سيصبح لدينا 4 متغيرات

A B C X

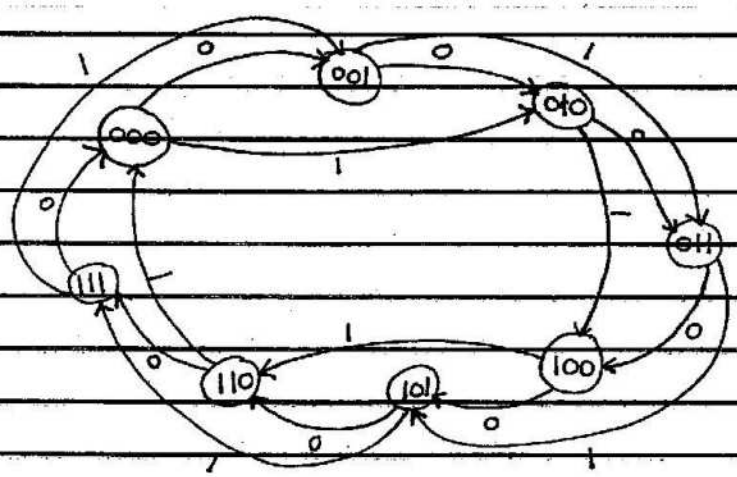
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state table

Present state A B C	input x	Next state A B C	FF-inputs excitation T_A T_B T_C
0 0 0	0	0 0 1	0 0 1
0 0 0	1	0 1 0	0 1 0
0 0 1	0	0 1 0	0 1 1
0 0 1	1	0 1 1	0 1 0
0 1 0	0	0 1 1	0 0 1
0 1 0	1	1 0 0	1 1 0
0 1 1	0	1 0 0	1 1 1
0 1 1	1	1 0 1	1 1 0
1 0 0	0	1 0 1	0 0 1
1 0 0	1	1 1 0	0 1 0
1 0 1	0	1 1 0	0 1 1
1 0 1	1	1 1 1	0 1 0
1 1 0	0	1 1 1	0 0 1
1 1 0	1	0 0 0	1 1 0
1 1 1	0	0 0 0	1 0 1
1 1 1	1	0 0 1	1 1 0

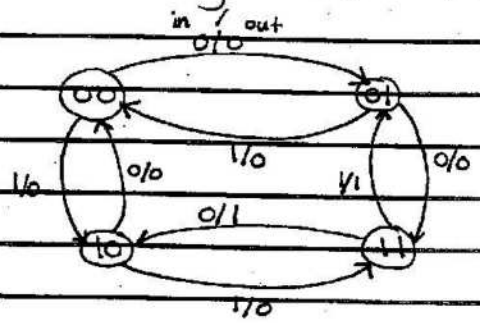
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state diagram



④ Given the state diagram below, generate the state table & design a sequential circuit using D FF.

PS	IN	NS	out
A B	X	A B	Y



ⓟ ← اولا بيبدأ تعبئة الجدول باحتمالات 3 متغيرات ثم نضيف اعمدة لـ DB و DA حتى نستطيع استكماله قيم واضافتها.

PS		IN	NS		out	FF inputs	
A	B	X	A	B	Y	D_A	D_B
0	0	0	0	1	0	0	1
0	0	1	1	0	0	1	0
0	1	0	1	1	0	1	1
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	0	1	1	0
1	1	1	0	1	1	0	1

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ⓠ ← ثانيا نأتي بـ NS و Y ونرجع للـ state diagram اللي فوق

ⓡ ← نأتي بـ D_A و D_B ونرجع للـ excitation table

ⓢ ← نحلهم بـ K-map

A \ BX	00	01	11	10
0		1		1
1		1		1

A \ BX	00	01	11	10
0	1			1
1		1	1	

A \ BX	00	01	11	10
0				
1			1	1

$D_A = \bar{B}X + B\bar{X}$

$= B \oplus X$

$D_B = \overline{\bar{A}\bar{X} + AX}$

$= \overline{A \oplus X}$

$Y = AB$

⑦

⑤ Design a 3-bits counter with sequence : $0 \to 3 \to 4 \to 6 \to 7$ and repeat (the unused states $\Rightarrow$ X).

Present state A B C	Next state A B C	FF-excitation J_A K_A	D_B	T_C
0 0 0	0 1 1	0 X	1	1
0 0 1	X X X	X X	X	X
0 1 0	X X X	X X	X	X
0 1 1	1 0 0	1 X	0	1
1 0 0	1 1 0	X 0	1	0
1 0 1	X X X	X X	X	X
1 1 0	1 1 1	X 0	1	1
1 1 1	0 0 0	X 1	0	1

unused states

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عند عمل الجدول : يجب تحديد A باستخدام موضع JK

D " " B

T " " C

لذلك نلتزم بال excitation table لكل موضع عند توجه

A \ BC	00	01	11	10
0		X	1	X
1	X	X	X	X

$J_A = B$

A \ BC	00	01	11	10
0	1	X		X
1	1	X		1

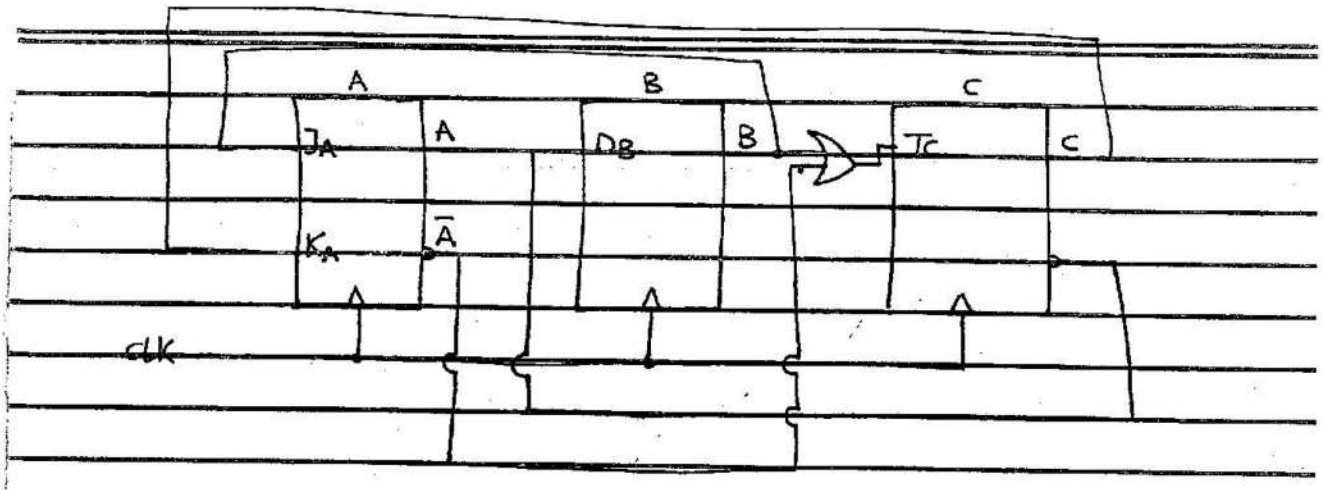
$D_B = \overline{C}$

A \ BC	00	01	11	10
0	X	X	X	X
1		X	1	

$K_A = C$

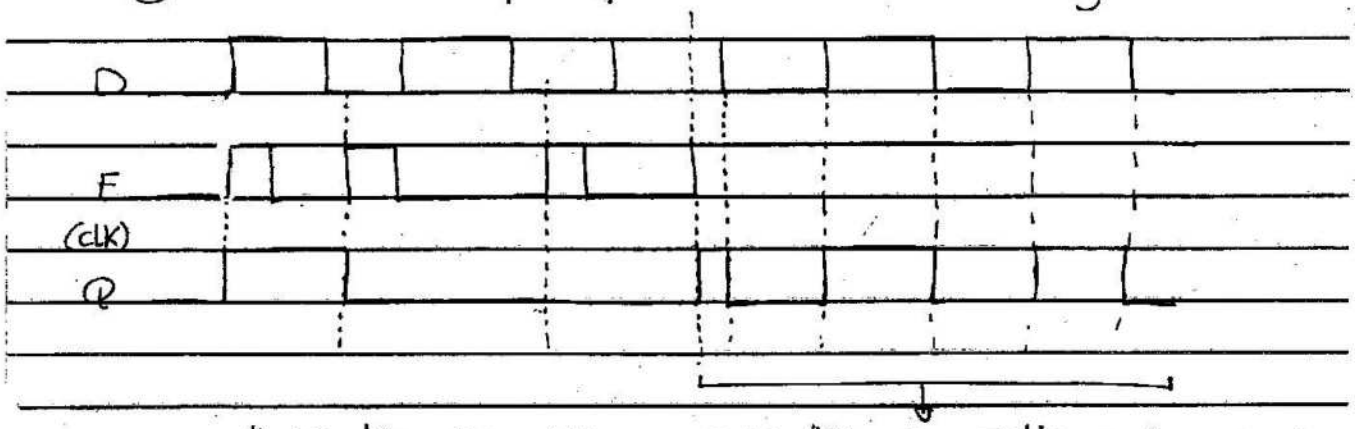
A \ BC	00	01	11	10
0	1	X	1	X
1		X	1	1

$T_C = \overline{A} + B$



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⑥ Draw the shape of Q for the following D-latch.



* في هذه المنطقة E دائماً" مرتفعة ← وبالتالي Q ستتغير تباعاً" مع تغير D

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(7) Consider that your are given the following char. equation

$$A(t+1) = x\bar{A}(t) \oplus \bar{B}(t)$$

$$B(t+1) = A(t) \oplus B(t)$$

$$Y(t) = A(t) + B(t) + x$$

using D.FF.

1- Fill the state table.

حيث A و B = مخرج فاف ، x = مدخل فاف ، Y = مخرج فاف.

Present state		input	Next state		out
A	B	X	A	B	Y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

↑ بالتعويض بالمعادلات المعطاة في السؤال

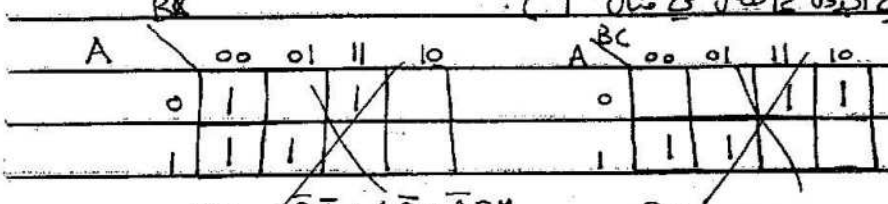
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2- Derive the input equations to FF.

$D_A = A(t+1)$ ← لانه المرجاع D فاف

$D_B = B(t+1)$

(ويكون على عمودين زيها في الجدول كما فصل في مثال 1).



$D_A = B\bar{X} + A\bar{B} + \bar{A}BX$ ، $D_B =$ ــــ

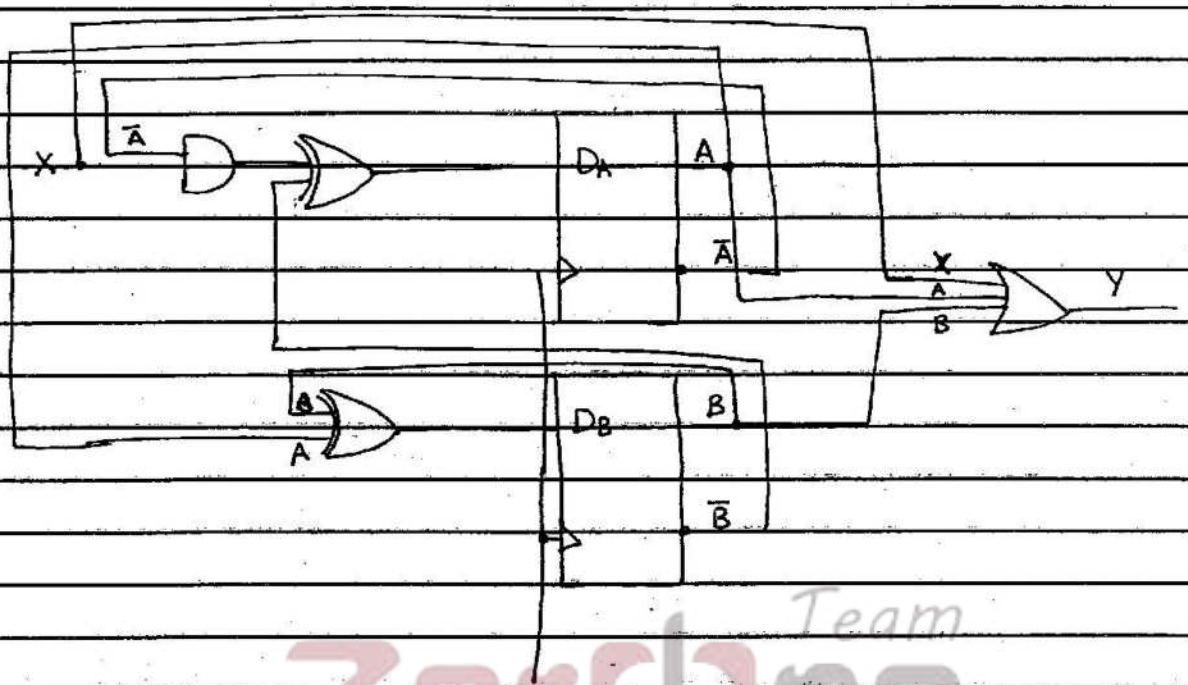
وبالتالي الاستغناء عن خطوة الاختصار ←

ورجع المخرج مباشرة حسب معادلة

$D = Q(t+1)$

لاننا وجد الاختصار سنحصل على نفس النتيجة

3- Draw the circuit diagram:-



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⑧ same example as ⑦ Design with SR FF.

$A(t+1) = X.\bar{A} \oplus \bar{B}$

$B(t+1) = A \oplus B$

$Y(t) = A + B + X.$

Present state A	B	inp X	Next state A	B	out Y	FF- inputs T_A	T_B
0	0	0	1	0	0	1	0
0	0	1	0	0	1	0	0
0	1	0	0	1	1	0	0
0	1	1	1	1	1	1	0
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	0	0	1	1	1
1	1	1	0	0	1	1	1

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$Q(t+1) = T \oplus Q(t)$ هنا نضع لقيمة T لاجل الوصول T

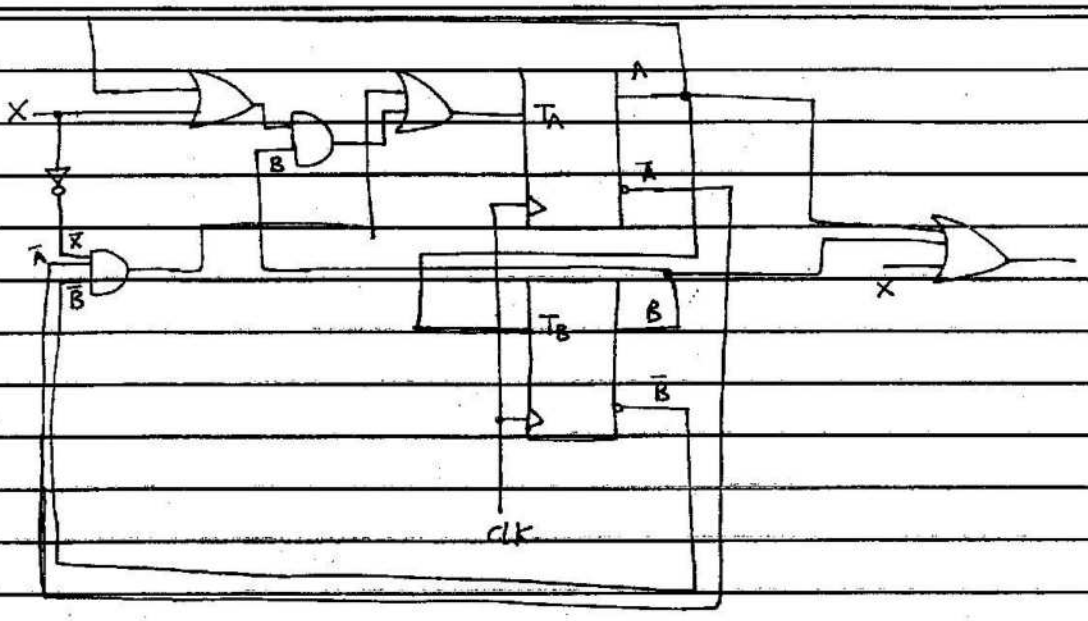
A \ BX	00	01	11	10
0	1		1	
1			1	1

$T_A = AB + BX + \bar{A}\bar{B}\bar{X}$

$B(A+X) + \bar{A}\bar{B}\bar{X}$

A \ BX	00	01	11	10
0				
1	1	1	1	1

$T_B = A$



⇐ قم بحل نفس المثال باستخدام :

JK (1

SR (2

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② Design the following counter - with unused states. using FF:

$8 \rightarrow 10 \rightarrow 12 \rightarrow 13 \rightarrow 15 \rightarrow 1 \rightarrow 8$

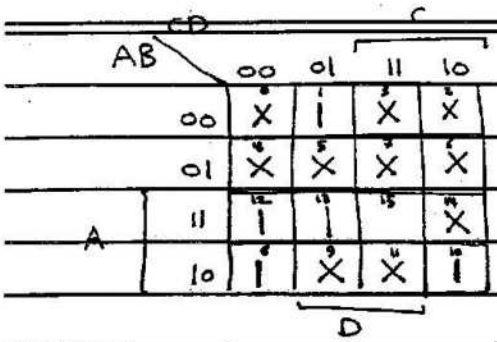
⇐ اعلى رقم 15 ← 4 bits بحاجة الى لتمثيله وبالتالي بحاجة الى 4 مراجع

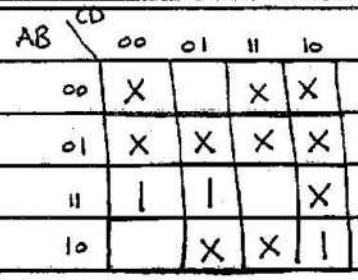
Present state	Next state	FF - inputs
A B C D	A B C D	D_A D_B D_C D_D
0 0 0 0	x x x x	x x x x
0 0 0 1	1 0 0 0	1 0 0 0
0 0 1 0	x x x x	x x x x
0 0 1 1	x x x x	x x x x
0 1 0 0	x x x x	x x x x
0 1 0 1	x x x x	x x x x
0 1 1 0	x x x x	x x x x
0 1 1 1	x x x x	x x x x
1 0 0 0	1 0 1 0	1 0 1 0
1 0 0 1	x x x x	x x x x
1 0 1 0	1 1 0 0	1 1 0 0
1 0 1 1	x x x x	x x x x
1 1 0 0	1 1 0 1	1 1 0 1
1 1 0 1	1 1 1 1	1 1 1 1
1 1 1 0	x x x x	x x x x
1 1 1 1	0 0 0 1	0 0 0 1

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dont care الحالات التي لم تظهر في العداد اعتبرها

0, 2, 3, 4, 5, 6, 7, 9, 11, 14





$D_A = \bar{D} + \bar{C}$

$D_B = B\bar{C} + C\bar{D}$

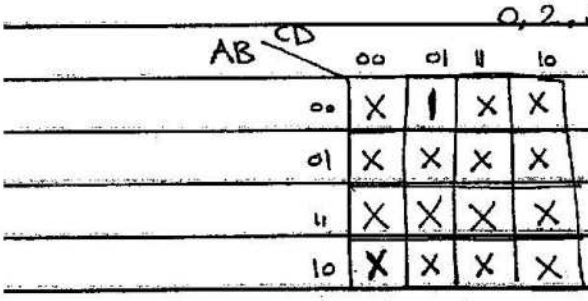
⇐ أكمل اختصار D_C و D_B.

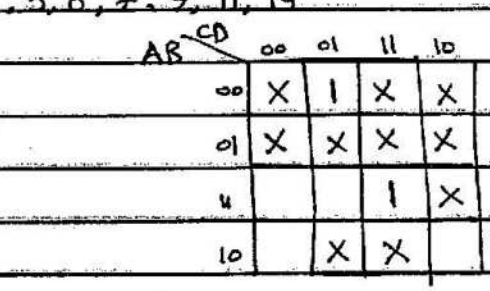
⇐ يمكنه حل السؤال بالرجوع لقيمة الخرج بالحالات المهملة (للسرعة) خاصة اذا طلب منك استخدام JK او SR.

Present state				Next state				FF- inputs					
A	B	C	D	A	B	C	D	J_A	K_A	J_B	K_B	J_C	K_C	J_D	K_D
1	0	0	0	1	0	1	0	X	0	0	X	1	X	0	X
1	0	1	0	1	1	0	0	X	0	1	X	X	1	0	X
1	1	0	0	1	1	0	1	X	0	X	0	X	0	1	X
1	1	0	1	1	1	1	1	X	0	X	0	1	X	X	0
1	1	1	1	0	0	0	1	X	1	X	1	X	1	X	0
0	0	0	1	1	0	0	0	1	X	0	X	0	X	X	1

⇐ لا داعي لوضع الحالات المهملة في الجدول (للسرعة) - لكن يجب ان تضعوا في الـ K-map - وهي ذات اعداد ثابتة

0, 2, 3, 4, 5, 6, 7, 9, 11, 14





$J_A = 1$

$K_A = CD$

⇐ وهكذا لباقي المراجع (أكمل الحل).

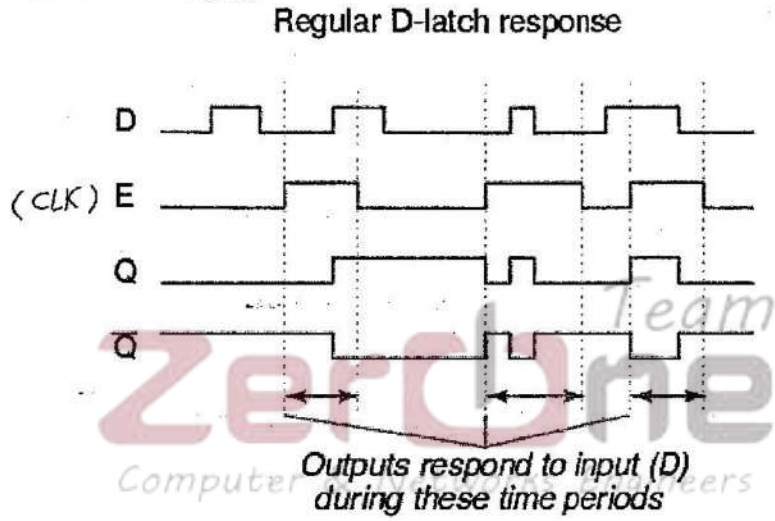
Part One: Latches and Flip-Flops

Clocked Latches:

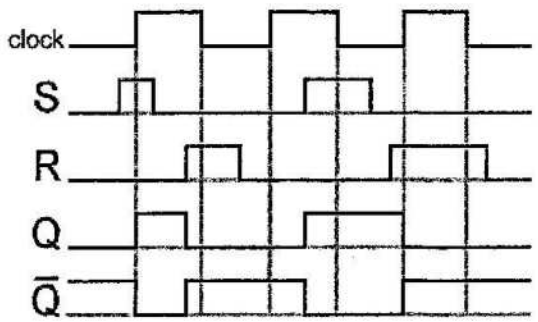
Latches are a level sensitive devices, they change their state only when the CLK (or E) is high.

Examples: Draw the shapes of Q and Q' for the following latches, given the shape of the input signals and the CLK (E):

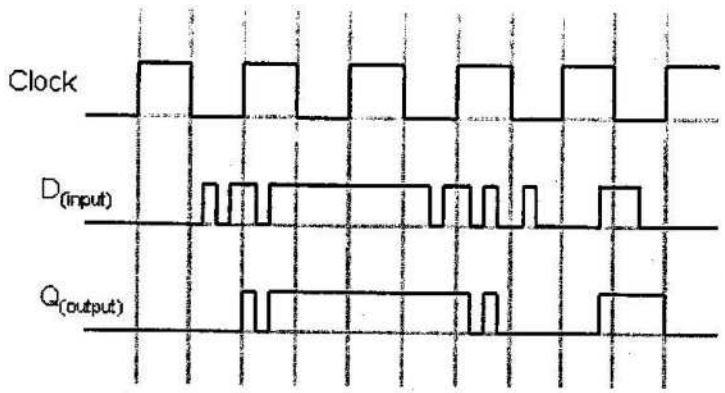
1. D latch: E is the clock (clk)



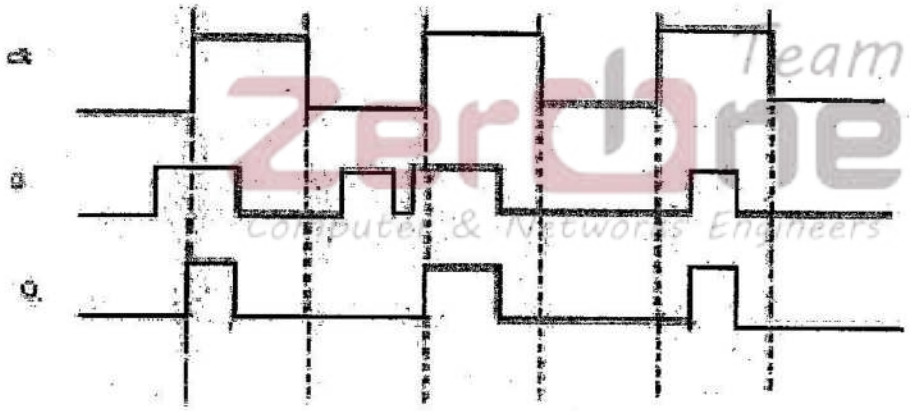
2. SR Latch:



3. D Latch:

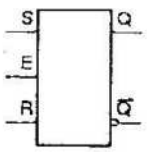


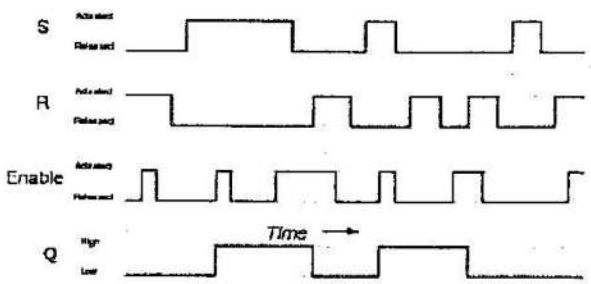
4. D latch



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5. SR Latch





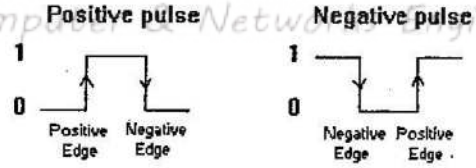
Clocked Flip-Flops:

Flip-Flops are edge triggered devices, they change their state at the edge of the clock. There are two types:

1- Positive edge FF (rising edge): changes the state of the FF when the clock transforms from low to high.

2- Negative edge FF (falling edge): changes the state of the FF when the clock transforms from high to low.

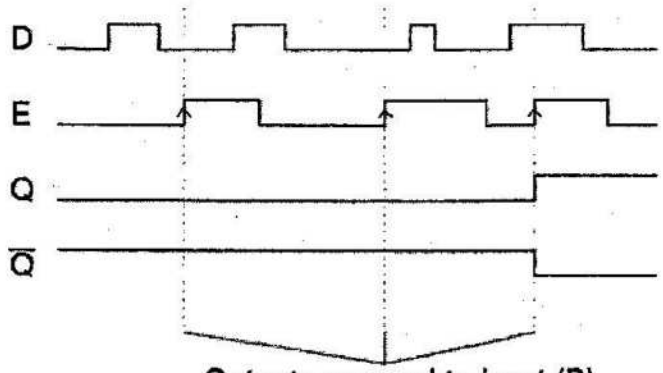
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Examples: Draw the shapes of Q and Q' for the following FF's, given the shape of the input signals and the CLK (E):

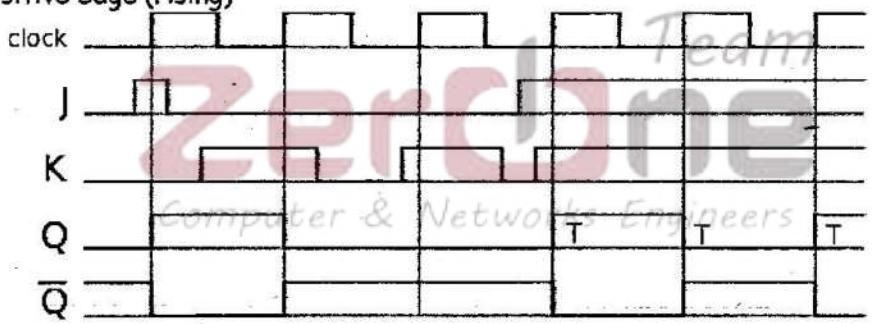
1) D FF - rising edge (E is the clock):

Positive edge-triggered D-latch response



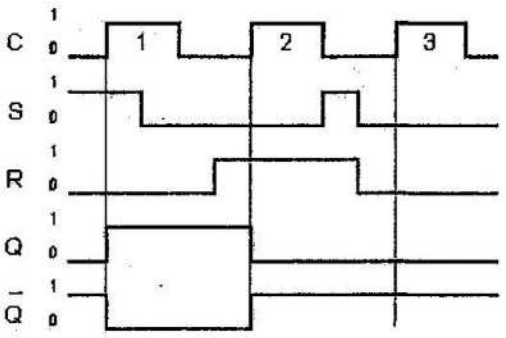
Outputs respond to input (D) only when enable signal transitions from low to high

2) JK FF - Positive edge (rising)

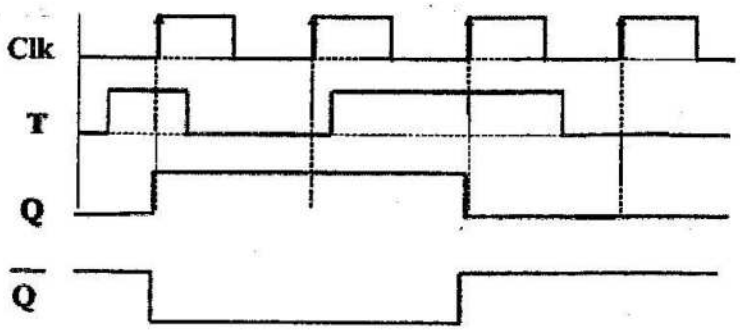


T = toggle

3) SR FF - rising edge:

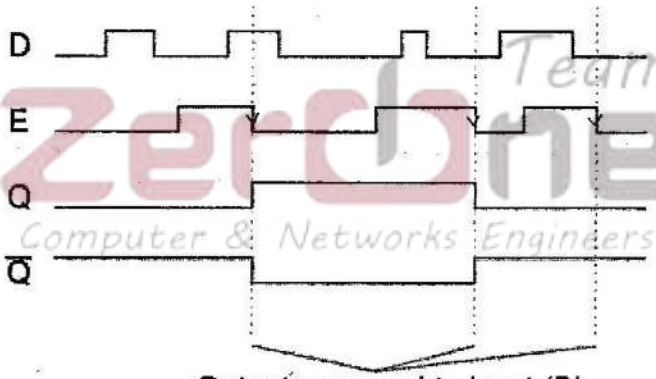


4) T FF - rising edge:



5) D FF - falling edge:

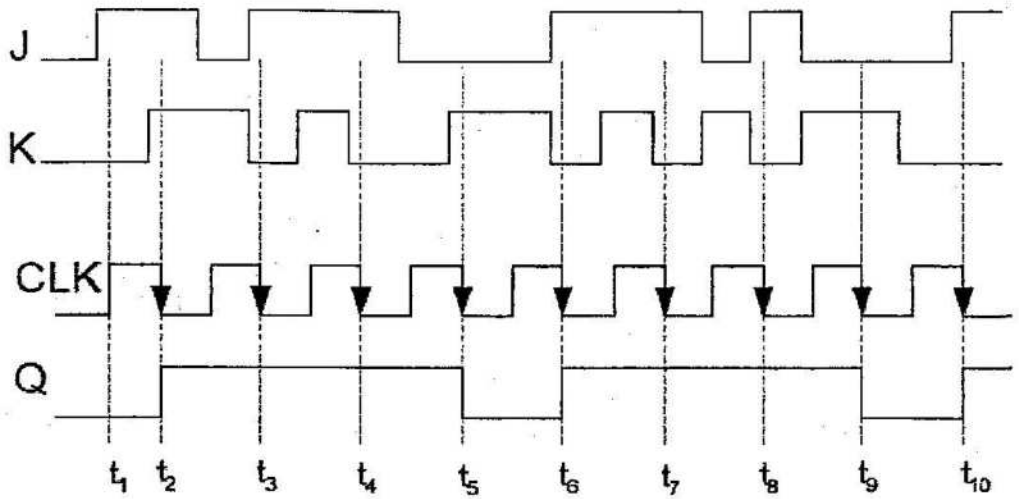
Negative edge-triggered D-latch response



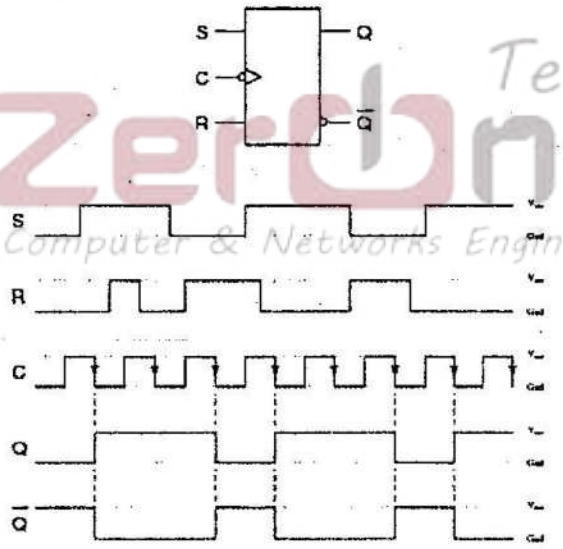
Outputs respond to input (D) only when enable signal transitions from high to low

6) JK Ff - negative edge:

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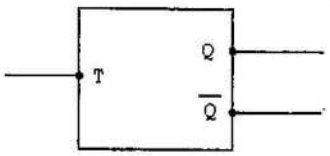
7) SR FF - negative edge:



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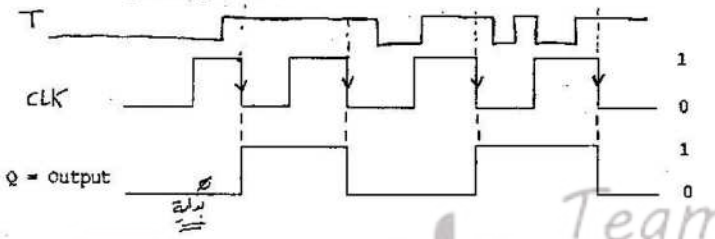
8) T FF - Falling edge:

T Flip - Flop

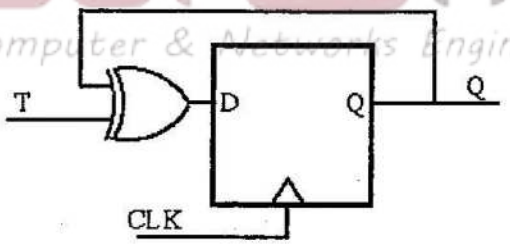


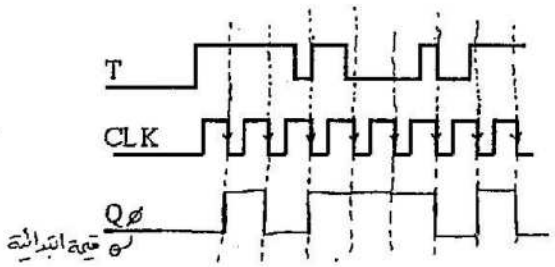
CLOCK	T	Q	Q'
↓	0	1	0
↓	1	0	0
↓	0	1	0
↓	1	0	1

Timing Diagram



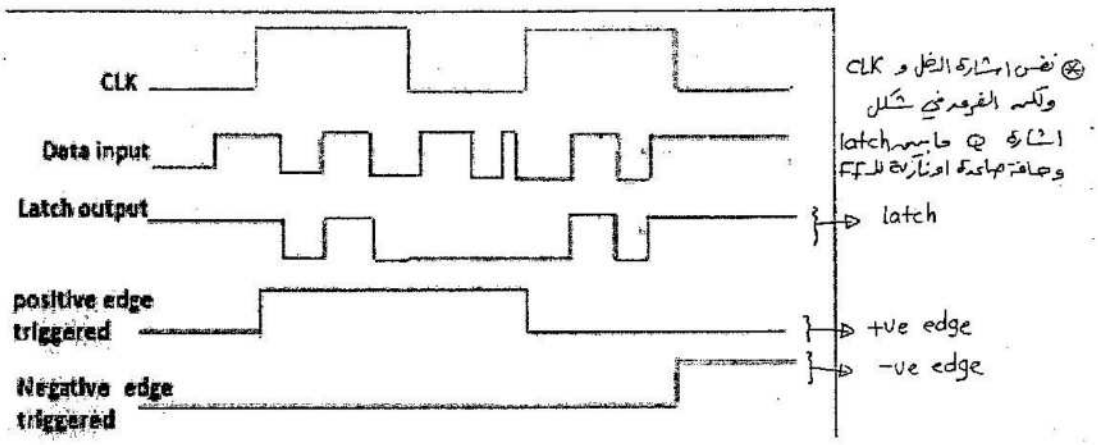
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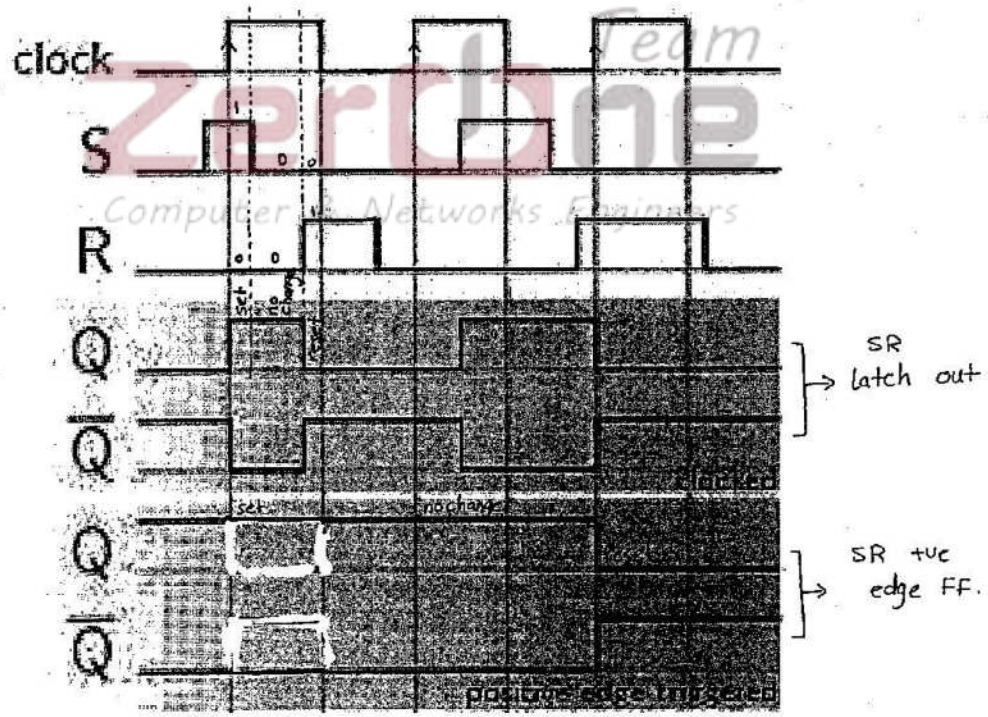
Example:

Difference between the output of a D latch and a D FF (positive and negative) for the same input and CLK.



Example 2:

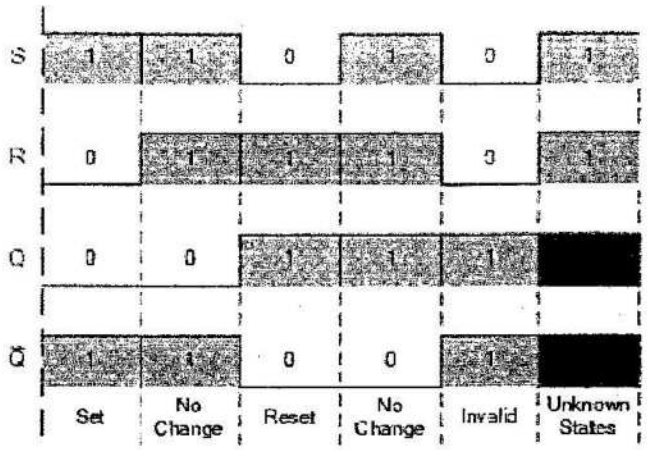
Difference between the output of a SR Latch and SR positive edge FF for the same input and CLK.



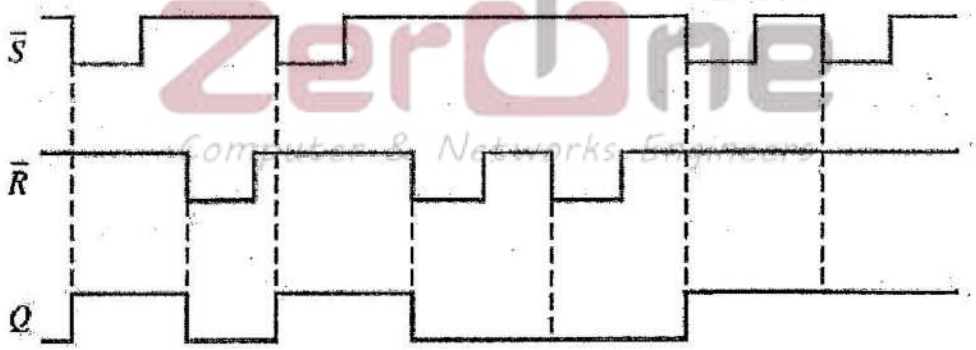
Flip Flops and Latches examples without clock:

Example: Draw the shape of Q (Q') if you are given the shapes of the inputs - no clock is considered.

1) SR:



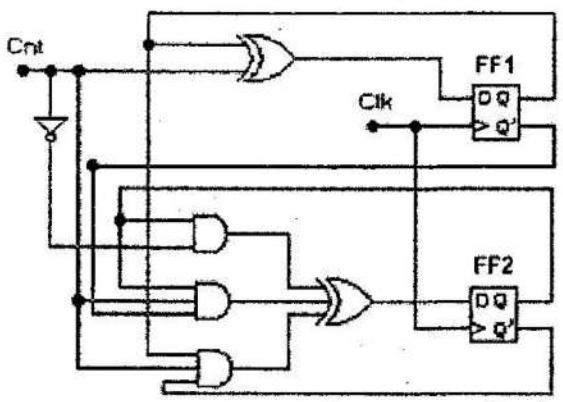
2) S'R':



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Part Two: Analysis of Sequential Circuits

Example 1: Derive the state table and state diagram for the sequential circuit shown below.



SOLUTION:

STEP 1: First we derive the Boolean expressions for the inputs of each flip-flops in the schematic, in terms of external input Cnt and the flip-flop outputs Q1 and Q0. Since there are two D flip-flops in this example, we derive two expressions for D1 and D0:

D0 = Cnt⊕Q0 = Cnt'*Q0 + Cnt*Q0'

D1 = Cnt'*Q1 + Cnt*Q1'*Q0 + Cnt*Q1*Q0'

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These Boolean expressions are called **excitation equations** since they represent the inputs to the flip-flops of the sequential circuit in the next clock cycle.

STEP 2: Derive the next-state equations by converting these excitation equations into flip-flop characteristic equations. In the case of D flip-flops, Q(next) = D. Therefore the next state equal the excitation equations.

Q0(next) = D0 = Cnt'*Q0 + Cnt*Q0'

Q1(next) = D1 = Cnt'*Q1 + Cnt*Q1'*Q0 + Cnt*Q1*Q0'

بالتعويض في المعادلات لتعبير Next state

STEP 3: Now convert these next-state equations into tabular form called the next-state table.

Present State	Next State	
Q1Q0	Cnt=0	Cnt=1
0 0	0 0	0 1
0 1	0 1	1 0
1 0	1 0	1 1
1 1	1 1	0 0

PS Q1	PS Q0	inp Cnt	NS Q1	NS Q0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

3 أسلوب اخر لتعبئة الجدول (التزم بأسلوب المحاضرات)

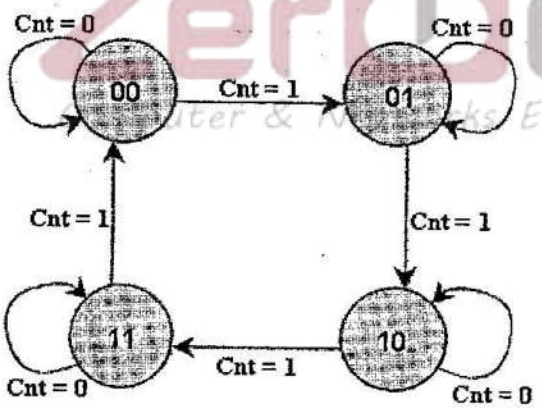
Each row is corresponding to a state of the sequential circuit and each column represents one set of input values. Since we have two flip-flops, the number of possible states is four - that is, Q1Q0 can be equal to 00, 01, 10, or 11. These are present states as shown in the table.

For the next state part of the table, each entry defines the value of the sequential circuit in the next clock cycle after the rising edge of the **Clk**. Since this value depends on the present state and the value of the input signals, the next state table will contain one column for each assignment of binary values to the input signals. In this example, since there is only one input signal, **Cnt**, the next-state table shown has only two columns, corresponding to **Cnt = 0 and Cnt = 1.**

Note that each entry in the next-state table indicates the values of the flip-flops in the next state if their value in the present state is in the row header and the input values in the column header.

Each of these next-state values has been computed from the next-state equations in STEP 2.

STEP 4: The state diagram is generated directly from the next-state table, shown below.



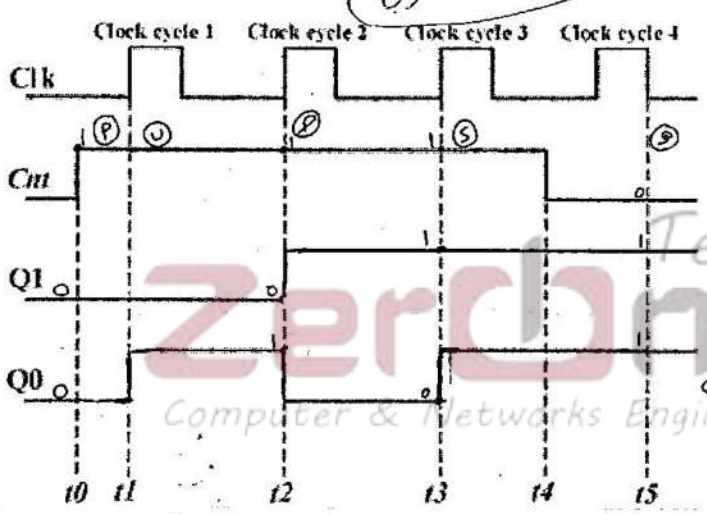
Each arc is labeled with the values of the input signals that cause the transition from the present state (the source of the arc) to the next state (the destination of the arc).

In general, the number of states in a next-state table or a state diagram will equal 2^m, where m is the number of flip-flops. Similarly, the number of arcs will equal $2^m \times 2^k$, where k is the number of binary input signals. Therefore, in the state diagram, there must be four states and eight transitions. Following these transition arcs, we can see that as long as Cnt = 1, the sequential circuit goes through the states in the following sequence: **0, 1, 2, 3, 0, 1, 2,**

.... On the other hand, when Cnt = 0, the circuit stays in its present state until Cnt changes to 1, at which the counting continues.

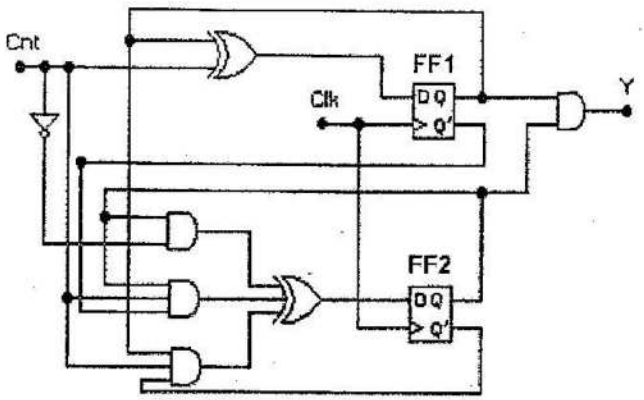
Since this sequence is characteristic of modulo-4 counting, we can conclude that the sequential circuit in Figure 7 is a modulo-4 counter with one control signal, Cnt, which enables counting when Cnt = 1 and disables it when Cnt = 0.

Below, we show a timing diagram, representing four clock cycles, which enables us to observe the behavior of the counter in greater detail.



In this timing diagram we have assumed that Cnt is asserted in clock cycle 0 at t_0 and is disasserted in clock cycle 3 at time t_4. We have also assumed that the counter is in state Q1Q0 = 00 in the clock cycle 0. Note that on the clock's rising edge, at t_1, the counter will go to state Q1Q0 = 01 with a slight propagation delay; in cycle 2, after t_2, to Q1Q0 = 10; and in cycle 3, after t_3 to Q1Q0 = 11. Since Cnt becomes 0 at t_4, we know that the counter will stay in state Q1Q0 = 11 in the next clock cycle.

Example 2: Derive the next state, the output table and the state diagram for the sequential circuit shown below.



نفس المثال السابق
مع وجود مخرج Y

SOLUTION:

The input combinational logic in Figure 10 is the same as in Example 1.1, so the excitation and the next-state equations will be the same as in Example 1.1.

Excitation equations:

$$D0 = Cnt \oplus Q0 = Cnt'*Q0 + Cnt*Q0'$$

$$D0 = Cnt'*Q1 + Cnt*Q1'*Q0 + Cnt*Q1*Q0'$$

Next-state equations:

$$Q0(next) = D0 = Cnt'*Q0 + Cnt*Q0'$$

$$Q1(next) = D0 = Cnt'*Q1 + Cnt*Q1'*Q0 + Cnt*Q1*Q0'$$

In addition, however, we have computed the output equation.

Output equation: $Y = Q1Q0$

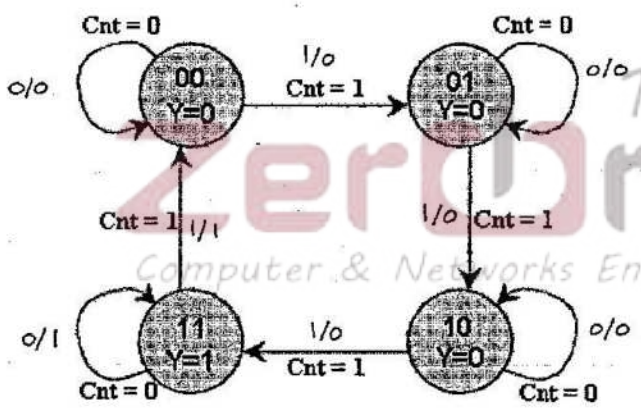
As this equation shows, the output Y will equal to 1 when the counter is in state Q1Q0 = 11, and it will stay 1 as long as the counter stays in that state.

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Next-state and output table:

Present State	Next State		Output
Q_1Q_0	Q_1Q_0	Cnt	Y
0 0	0 0	0 1	0
0 1	0 1	1 0	0
1 0	1 0	1 1	0
1 1	1 1	0 0	1

State diagram:

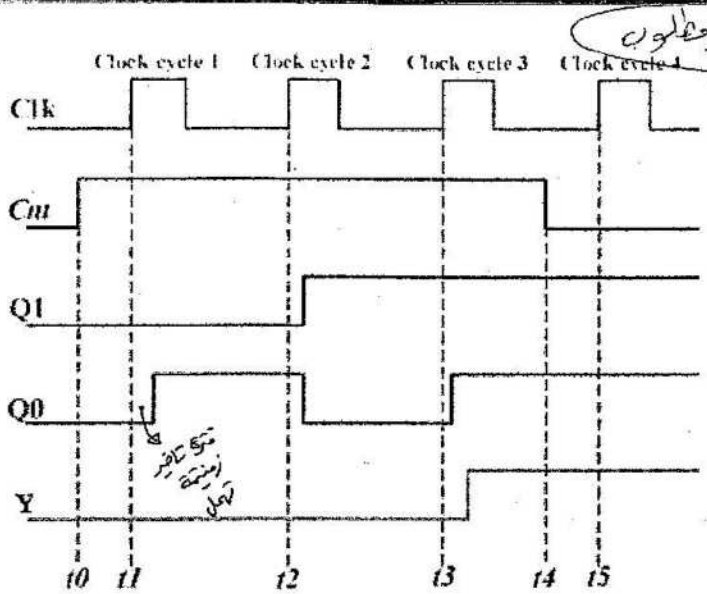


state table

Timing diagram:

PS Q_1	PS Q_0	inp Cnt	NS Q_1	NS Q_0	out Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

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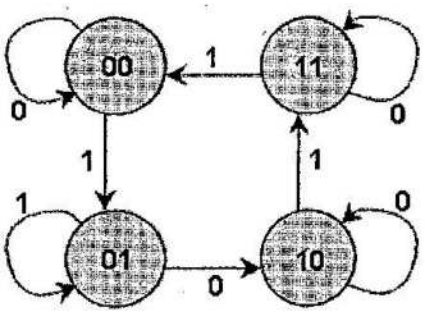


- تحويل الجدول او الـ state diagram لرسمة اشارات :-
- نفس شرح الرسمة السابقة مع وجود اشارة Y .
- يوجد تاخير في انتقال الاشارة الى القيم الجديدة (قم باهمال هذه الفترة)

Note that the counter will reach the state Q1Q0 = 11 only in the third clock cycle, so the output Y will equal 1 after Q0 changes to 1. Since counting is disabled in the third clock cycle, the counter will stay in the state Q1Q0 = 11 and Y will stay asserted in all succeeding clock cycles until counting is enabled again.

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Example 3: We wish to design a synchronous sequential circuit whose state diagram is shown below. The type of flip-flop to be use is J-K.



From the state diagram, we can generate the state table shown below. Note that there is no output section for this circuit. Two flip-flops are needed to represent the four states and are designated Q0Q1. The input variable is labeled x.

State Table

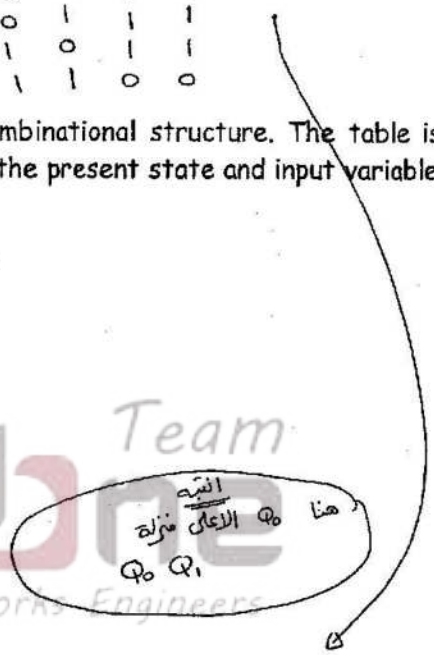
Present State	Next State	
Q_0 Q_1		
0 0	0 0	0 1
0 1	1 0	0 1
1 0	1 0	1 1
1 1	1 1	0 0

PS Q_0Q_1	inp x	NS Q_0 Q_1
0 0	0	0 0
0 0	1	0 1
0 1	0	1 0
0 1	1	0 1
1 0	0	1 0
1 0	1	1 1
1 1	0	1 1
1 1	1	0 0

We shall now derive the excitation table and the combinational structure. The table is now arranged in a different form shown in Table 11, where the present state and input variables are arranged in the form of a truth table.

Table 10. Excitation table for JK flip-flop

Output Transition	Flip-flop Inputs
Q_t Q_{t+1}	J K
0 0	0 X
0 1	1 X
1 0	X 1
1 1	X 0



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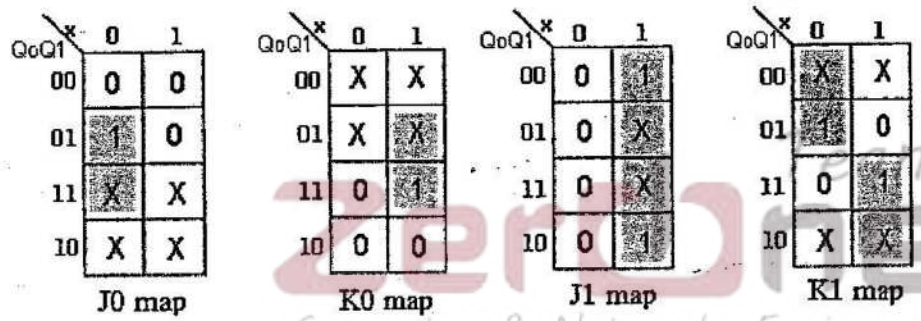
Table 11. Excitation table of the circuit

Present State	Next State	Input	Flip-flop Inputs	
Q_0 Q_1	Q_0 Q_1	x	J_0 K_0	J_1 K_1
0 0	0 0	0	0 X	0 X
0 0	0 1	1	0 X	1 X
0 1	1 0	0	1 X	X 1
0 1	0 1	1	0 X	X 0
1 0	1 0	0	X 0	0 X
1 0	1 1	1	X 0	1 X
1 1	1 1	0	X 0	X 0
1 1	0 0	1	X 1	X 1

نقوم باضافة احدى الـ JK لكل FF حتى نتعرف متى المطلوب كل وجاج - ونبتدئم حسب الـ excitation table

In the first row of Table 11, we have a transition for flip-flop Q0 from 0 in the present state to 0 in the next state. In Table 10 we find that a transition of states from 0 to 0 requires that input J = 0 and input K = X. So 0 and X are copied in the first row under J0 and K0 respectively. Since the first row also shows a transition for the flip-flop Q1 from 0 in the present state to 0 in the next state, 0 and X are copied in the first row under J1 and K1. This process is continued for each row of the table and for each flip-flop, with the input conditions as specified in Table 10.

The simplified Boolean functions for the combinational circuit can now be derived. The input variables are Q0, Q1, and x; the output are the variables J0, K0, J1 and K1. The information from the truth table is plotted on the Karnaugh maps shown in Figure 14.



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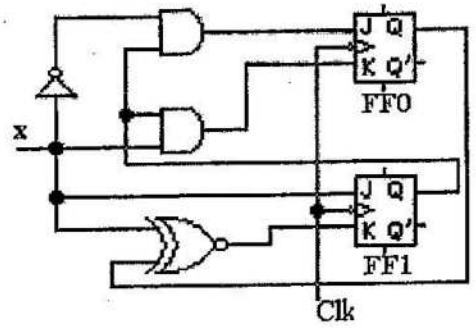
تتحقق باستخدام K-map.

The flip-flop input functions are derived:

J0 = Q1*x' **K0 = Q1*x**

J1 = x **K1 = Q0'*x' + Q0*x = Q0 XNOR x**

The logic diagram is drawn below.



نرسم الدائرة المنطقية

Example 4:

سؤال معطى فيه الـstate table ⇦

Design a sequential circuit whose state tables are specified in Table 12, using D flip-flops.

Table 12. State table of a sequential circuit.

00	00	01	0 0
01	00	10	0 0
10	11	10	0 0
11	00	01	0 1

PS Q_0Q_1	inp. x	NS Q_0Q_1	out Y
0 0	0	0 0	0
0 0	1	0 1	0
0 1	0	0 0	0
0 1	1	1 0	0
1 0	0	1 1	0
1 0	1	1 0	0
1 1	0	0 0	0
1 1	1	0 1	1

Table 13. Excitation table for a D flip-flop.

0 0	0
0 1	1
1 0	0
1 1	1

Q_0 Q_1
↓
MSB

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Next step is to derive the excitation table for the design circuit, which is shown in Table 14. The output of the circuit is labeled Z.

0 0	0 0	0	0	0	0
0 0	0 1	1	0	1	0

0 1	0 0	0	0	0	0
0 1	1 0	1	1	0	0
1 0	1 1	0	1	1	0
1 0	1 0	1	1	0	0
1 1	0 0	0	0	0	0
1 1	0 1	1	0	1	1

Table 14. Excitation table

قم بإضافة أسطر جديدة على الجدول المطلق
للداخل D_{Q_1} D_{Q_0}

Now plot the flip-flop inputs and output functions on the Karnaugh map to derive the Boolean expressions, which is shown in Figure 16.

QoQ1 \ x	0	1
00	0	0
01	0	1
11	0	0
10	1	1

D0 map

QoQ1 \ x	0	1
00	0	1
01	0	0
11	0	1
10	1	0

D1 map

QoQ1 \ x	0	1
00	0	0
01	0	0
11	0	1
10	0	0

Z map

ps Q_0Q_1	inp x	NS Q_0Q_1	out Y	FF-inputs D_0 D_1
0 0	0	0 0	0	0 0
0 0	1	0 1	0	0 1
0 1	0	0 0	0	0 0
0 1	1	1 0	0	1 0
1 0	0	1 1	0	1 1
1 0	1	1 0	0	1 0
1 1	0	0 0	0	0 0
1 1	1	0 1	1	0 1

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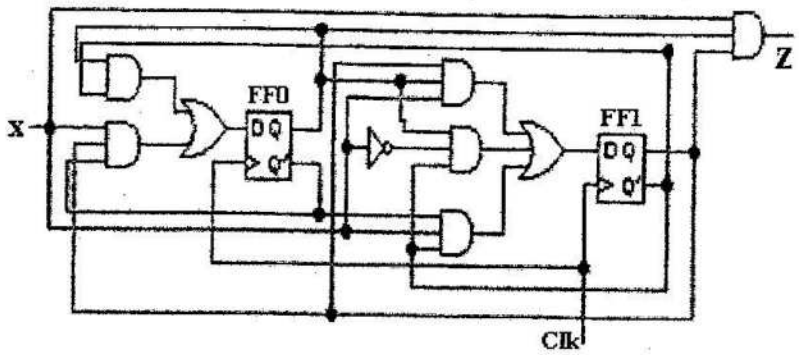
The simplified Boolean expressions are:

$D0 = Q0*Q1' + Q0'*Q1*x$

$D1 = Q0'*Q1'*x + Q0*Q1*x + Q0*Q1'*x'$

$Z = Q0*Q1*x$

Finally, draw the logic diagram.



Part Three: Synchronous Binary Counters

Example 1: A counter is first described by a state diagram, which is shows the sequence of states through which the counter advances when it is clocked. Figure 18 shows a state diagram of a 3-bit binary counter.

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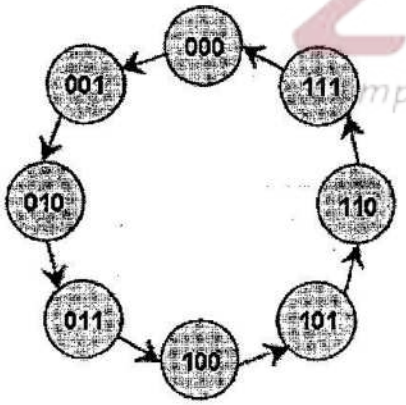


Figure 18. State diagram of a 3-bit binary counter.

The circuit has no inputs other than the clock pulse and no outputs other than its internal state (outputs are taken off each flip-flop in the counter). The next state of the counter depends entirely on its present state, and the state transition occurs every time the clock pulse occurs. Figure 19 shows the sequences of count after each clock pulse.

Once the sequential circuit is defined by the state diagram, the next step is to obtain the next-state table, which is derived from the state diagram in Figure 18 and is shown in Table 15.

Table 15. State table

Present State	Next State
Q2 Q1 Q0	Q2 Q1 Q0
0 0 0	0 0 1
0 0 1	0 1 0
0 1 0	0 1 1
0 1 1	1 0 0
1 0 0	1 0 1
1 0 1	1 1 0
1 1 0	1 1 1
1 1 1	0 0 0

Since there are eight states, the number of flip-flops required would be three. Now we want to implement the counter design using JK flip-flops.

Next step is to develop an excitation table from the state table, which is shown in Table 16.

Table 16. Excitation table

Output State Transitions		Flip-flop Inputs		
Present State	Next State			
Q2 Q1 Q0	Q2 Q1 Q0	J2 K2	J1 K1	J0 K0
0 0 0	0 0 1	0 X	0 X	1 X
0 0 1	0 1 0	0 X	1 X	X 1
0 1 0	0 1 1	0 X	X 0	1 X
0 1 1	1 0 0	1 X	X 1	X 1
1 0 0	1 0 1	X 0	0 X	1 X
1 0 1	1 1 0	X 0	1 X	X 1
1 1 0	1 1 1	X 0	X 0	1 X
1 1 1	0 0 0	X 1	X 1	X 1

Now transfer the JK states of the flip-flop inputs from the excitation table to Karnaugh maps to derive a simplified Boolean expression for each flip-flop input. This is shown in Figure 20.

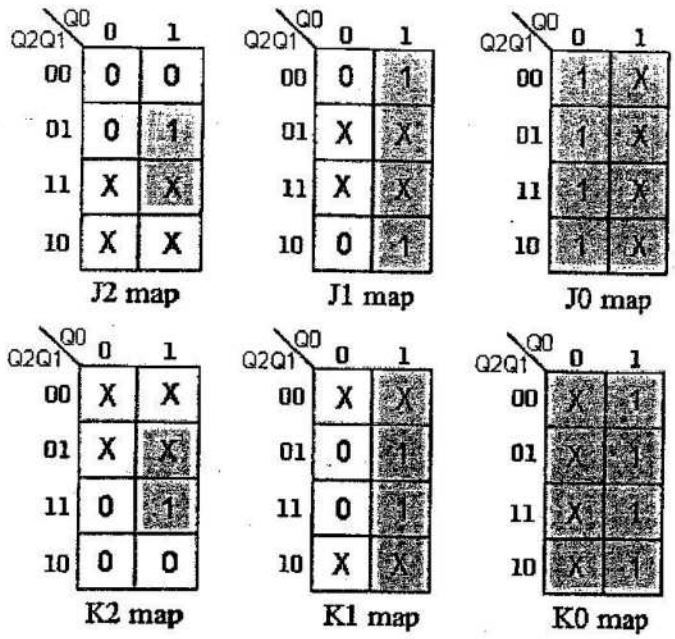


Figure 20. Karnaugh maps

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The 1s in the Karnaugh maps of Figure 20 are grouped with "don't cares" and the following expressions for the J and K inputs of each flip-flop are obtained:

J0 = K0 = 1

J1 = K1 = Q0

J2 = K2 = Q1*Q0

The final step is to implement the combinational logic from the equations and connect the flip-flops to form the sequential circuit. The complete logic of a 3-bit binary counter is shown in Figure 21.

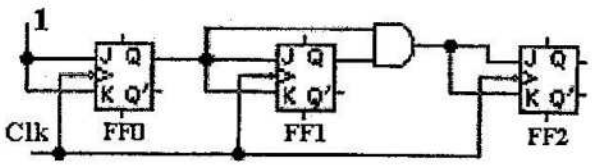


Figure 21. Logic diagram of a 3-bit binary counter.

Example 2:

Design a counter specified by the following state diagram using T flip-flops. The state diagram is shown here again in Figure 22.

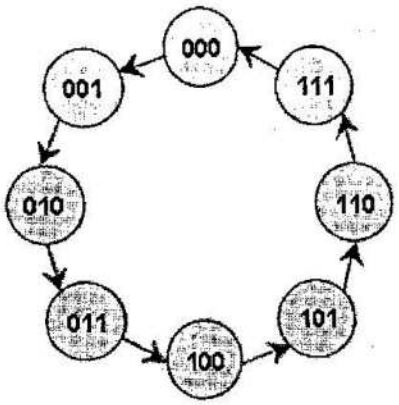


Figure 22. State diagram of a 3-bit binary counter.

The state table will be the same as in Example 1.

Now derive the excitation table from the state table, which is shown in Table 17.

Table 17. Excitation table.

Output State Transitions		Flip-flop Inputs
Present State	Next State	
$Q_2\ Q_1\ Q_0$	$Q_2\ Q_1\ Q_0$	$T_2\ T_1\ T_0$
0 0 0	0 0 1	0 0 1
0 0 1	0 1 0	0 1 1
0 1 0	0 1 1	0 0 1
0 1 1	1 0 0	1 1 1
1 0 0	1 0 1	0 0 1
1 0 1	1 1 0	0 1 1
1 1 0	1 1 1	0 0 1
1 1 1	0 0 0	1 1 1

Next step is to transfer the flip-flop input functions to Karnaugh maps to derive a simplified Boolean expressions, which is shown in Figure 23.

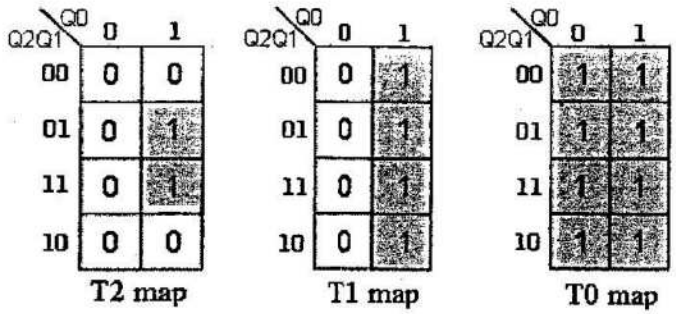


Figure 23. Karnaugh maps

The following expressions are obtained:

T0 = 1; T1 = Q0; T2 = Q1*Q0

Finally, draw the logic diagram of the circuit from the expressions obtained. The complete logic diagram of the counter is shown in Figure 24.

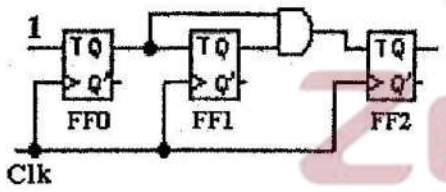


Figure 24. Logic diagram of 3-bit binary counter.

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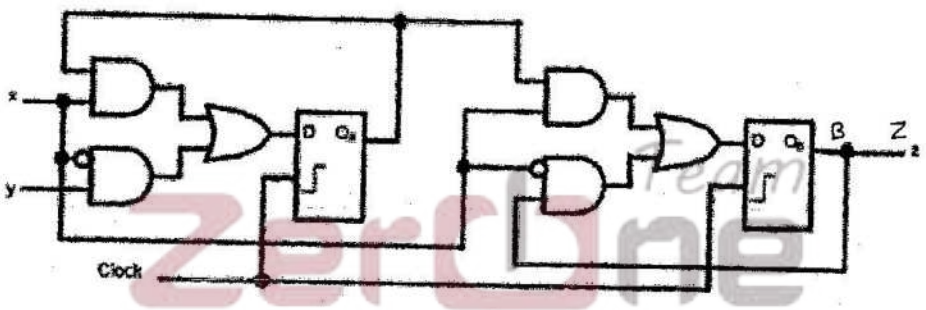
Example 3: A sequential circuit uses two D flip-flops (A and B) as memory elements. The behavior of the circuit is described by the following equations:

$A(t+1) = x'y + xA$

$B(t+1) = x'B + xA$

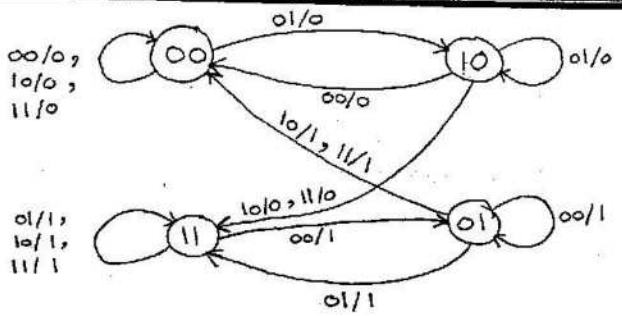
$z = B$

Draw the logic diagram of the circuit then derive the state table and draw the state diagram of the circuit.



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State		Input		Next state		Output
A	B	x	y	A	B	z
0	0	0	0	0	0	0
0	0	0	1	1	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	0	1	1	0
1	0	1	1	1	1	0
1	1	0	0	0	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1



Exercises

You can try some of these exercises which covers the analysis and design of sequential circuits.

Analysis of Sequential Circuits.

1. Derive a) excitation equations, b) next state equations, c) a state/output table, and d) a state diagram for the circuit shown in Figure 1.1. Draw the timing diagram of the circuit.

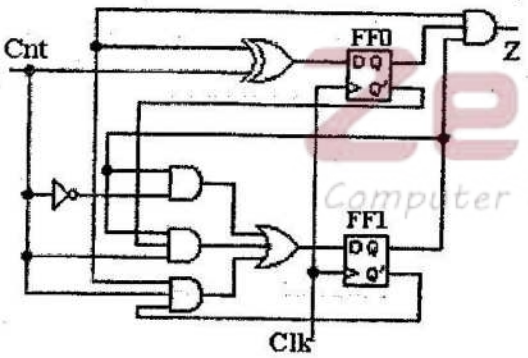


Figure 1.1

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2. Derive a) excitation equations, b) next state equations, c) a state/output table, and d) a state diagram for the circuit shown in Figure 1.2.

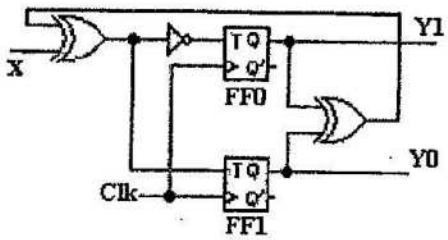


Figure 1.2

Design of Sequential Circuits.

6. Design a sequential circuit specified by Table 6.1, using JK flip-flops.

Table 6.1

Present State	Next State		Output	
Q_0Q_1	$x=0$	$x=1$	$x=0$	$x=1$
00	00	01	0	0
01	00	10	0	0
10	11	10	0	0
11	00	01	0	1

PS Q_0Q_1	in x	NS Q_0Q_1	out y
0 0	0	0 0	0
0 0	1	0 1	0
0 1	0	0 0	0
0 1	1	1 0	0
1 0	0	1 1	0
1 0	1	1 0	0
1 1	0	0 0	0
1 1	1	0 1	1

7. Design the sequential circuit in question 6, using T flip-flops.

8. Design a mod-5 counter which has the following binary sequence: 0, 1, 2, 3, 4. Use JK flip-flops.

9. Design a counter that has the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6, 7. Use RS flip-flops.

10. Design a counter with the following binary sequence: 1, 2, 5, 7 and repeat. Use JK flip-flops.

11. Design a counter with the following repeated binary sequence: 0, 4, 2, 1, 6. Use T flip-flops.

12. Design a counter that counts in the sequence 0, 1, 3, 6, 10, 15, using four a) D, b) SR, c) JK and d) T flip-flops.

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